

# Osaid Khan

+1 (945) 304-5781 | [khanosaid726@gmail.com](mailto:khanosaid726@gmail.com) | [www.linkedin.com/in/-osaid-khan/](https://www.linkedin.com/in/-osaid-khan/) | [github.com/ozzyozbourne](https://github.com/ozzyozbourne)

## EDUCATION

### Pace University

*Master of Science (MS) in Computer Science (GPA - 4.0/4.0)*

New York City, NY

*Jan 2024 - Dec 2025*

### Dr. A.P.J. Abdul Kalam Technical University

*Bachelor of Technology (B.Tech) in Computer Science*

Noida, India

*Jul 2016 - Sep 2020*

## TECHNICAL SKILLS

**AWS Services:** EC2, RDS, S3, Lambda, CloudWatch, Step Functions, CloudFormation, CloudFront, API Gateway

**Languages:** Java, Python, JavaScript/TypeScript, Rust, Html/Css

**Backend & Frameworks:** Spring Boot, FastAPI, LangGraph, LangChain, FastMCP, React, LangSmith, Langfuse

**Databases:** PostgreSQL, Redis, MongoDB

**Others:** Git, CI/CD, Docker, Kubernetes, OpenAI Agents SDK, LlamaIndex, Google ADK, Claude Code

## PROFESSIONAL EXPERIENCE

### Software Engineer

Phoenix, AZ

Sperse

*Feb 2026 - Present*

- Designed and implemented a **multi-agent** CRM platform in **Python**, **LangGraph**, and **FastAPI** serving 10,000+ users, routing requests across 40+ specialized sub-agents via a **distributed systems** task-routing architecture with tenant-aware access control — a **complex software system** engineered for reliability and horizontal scaling
- Engineered a **MCP** server using **FastMCP** exposing 160+ LLM-callable backend actions over a **multi-tiered** service, with built-in monitoring and Pydantic validation to ensure secure, reliable execution of **cloud computing** workflows across multiple customers
- Built **GitHub Actions CI/CD** pipelines for AI-agent deployments, integrating **OpenTelemetry** trace-based eval suites that catch latency regressions, accuracy degradation, and safety issues before production — improving release quality through automated design review
- Built **React Flow** and **TypeScript** visual workflow platform for creating and managing **distributed** agentic swarms, using **Temporal** for durable, resumable **multi-threaded** execution with fault-tolerant state management
- Built **LlamaIndex**-powered **RAG** agents using OpenAI embeddings and **pgvector**, identifying and resolving accuracy bottlenecks through iterative retrieval tuning to generate ready-to-run REST API examples and reduce support load

### Software Engineer

Noida, India

Qualitest

*Feb 2021 - Dec 2023*

- Designed a cloud-agnostic Storage Service in **Java** and **Spring Boot** with **PostgreSQL** and **Redis** handling file operations across **AWS S3**, **Google Cloud Storage**, and **MinIO** via expiring signed URLs — a **multi-tiered, distributed** backend system built for cross-cloud **reliability and scaling**
- Orchestrated long-running jobs with **AWS Step Functions**, eliminating **Lambda** timeout failures and improving error recovery across **100K+** weekly executions — **architecture** and **design patterns** focused on fault tolerance and scaling of **complex software systems**
- Identified and resolved **performance bottlenecks** by optimizing **PostgreSQL** with window functions, materialized views, and indexes, then exposed results via interactive **React** dashboards for business stakeholders

## PROJECTS

### Dark Factory - orchestrator for PI agent | *Elixir, TypeScript*

[github.com/ozzyozbourne/dark-factory](https://github.com/ozzyozbourne/dark-factory)

- Built a **distributed systems** coding-agent orchestrator in **Elixir** and **TypeScript** that decomposes large requests into dependency-aware task DAGs and executes them across multi-machine AI agents with retries, timeout handling, and fault-tolerant **multi-threaded** state management